Blockchain Integrated Machine Learning System for Cryptocurrency Fraud Detection

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Abstract

Cryptocurrency platforms process millions of irreversible transactions daily, making fraud detection a critical challenge. Traditional fraud detection systems rely on centralized infrastructures, which are vulnerable to tampering and lack transparency. Machine learning techniques can detect fraud patterns effectively, but their decisions are often opaque and difficult to audit.

This project presents a hybrid framework that integrates machine learning with blockchain technology to build a transparent and tamper-proof cryptocurrency fraud detection system. A Random Forest classifier is trained on real Ethereum transaction data to identify fraudulent behavior. Each prediction is cryptographically hashed and permanently stored on the Ethereum Sepolia test network using smart contracts. This approach ensures that fraud detection decisions are immutable, verifiable, and trustworthy.

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Chapter 1

Introduction

Blockchain-based cryptocurrencies such as Bitcoin and Ethereum have revolutionized digital finance by enabling decentralized, peer-to-peer transactions without intermediaries. However, the absence of centralized control and the irreversible nature of blockchain transactions make these systems attractive targets for fraud.

Conventional fraud detection systems typically rely on centralized databases and rule-based mechanisms. Such systems suffer from a lack of transparency, single points of failure, and vulnerability to manipulation. Machine learning has shown strong performance in identifying fraudulent behavior, but the lack of interpretability and auditability reduces trust.

Blockchain provides immutability, decentralization, and public verification. This project integrates machine learning-based fraud detection with blockchain storage to ensure that every decision is permanently recorded and cannot be altered.

Chapter 2

Background and Related Work

Fraud detection has been extensively studied using statistical and machine learning approaches such as logistic regression, decision trees, and ensemble learning. Dal Pozzolo et al. demonstrated the importance of handling class imbalance in fraud detection problems.

Recent studies have explored deep learning techniques for financial fraud detection, but these methods require large datasets and significant computational resources. In the cryptocurrency domain, transaction graph analysis and address clustering have been used to identify illicit behavior.

Blockchain-based systems have been proposed for secure data storage and auditing, but they generally lack intelligent analysis. This project bridges this gap by integrating machine learning predictions directly with blockchain immutability.

Chapter 3

Problem Formulation

Each Ethereum transaction is represented by a feature vector:

$$\mathbf{x} = [b, t, v]$$

where b is the block height, t is the transaction timestamp, and v is the transaction value.

The objective is to learn a classifier:

$$f(\mathbf{x}) \rightarrow y, \quad y \in \{0, 1\}$$

where $y = 1$ denotes a fraudulent transaction and $y = 0$ denotes a legitimate transaction.

In addition to prediction accuracy, the system must satisfy:

- Immutability of predictions
- Public verifiability
- Near real-time operation

Chapter 4

Dataset and Preprocessing

The dataset consists of 254,973 real Ethereum transactions. Each transaction contains metadata such as transaction hash, block height, timestamp, sender address, receiver address, value, and an error flag.

The `isError` attribute is used as the target label. Address fields and transaction hashes are removed as they do not contribute to predictive learning. Only 6.13% of the transactions are fraudulent, resulting in a highly imbalanced dataset.

Class-weighted learning is used to address imbalance. Missing values are replaced with zeros, and non-informative features are removed.

Chapter 5

Machine Learning Model

Fraud detection is formulated as a binary classification task. Three models were evaluated: Logistic Regression, Random Forest, and XGBoost.

Random Forest was selected due to its robustness and ensemble nature. It consists of multiple decision trees trained on random subsets of data and features. The final prediction is obtained through majority voting:

$$\hat{y} = \text{mode}\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_n(\mathbf{x})\}$$

The model achieved an F1-score of 0.8163, indicating strong performance on imbalanced data.

Chapter 6

Fraud Detection Algorithm

Algorithm 1 Blockchain-based Fraud Detection

1: Input transaction features $\mathbf{x}$
2: Predict fraud label using Random Forest
3: Generate SHA-256 hash of $\mathbf{x}$
4: Store hash, label, and confidence on blockchain
5: Return prediction to user

Chapter 7

Blockchain Design

Blockchain ensures that fraud predictions cannot be modified. Each transaction input is hashed using SHA-256. The smart contract stores:

- Transaction hash
- Fraud label
- Confidence score
- Model identifier

The contract is deployed on the Ethereum Sepolia test network, ensuring decentralization and public verification.

Chapter 8

System Architecture

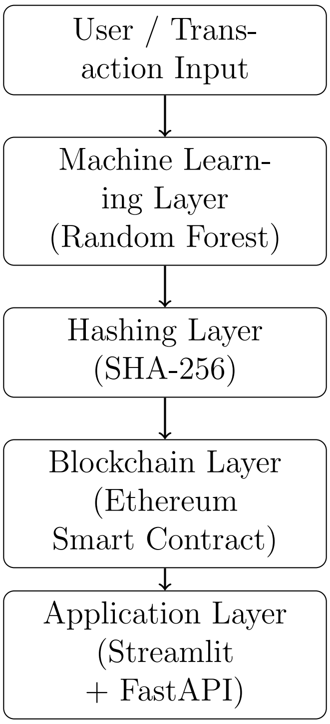

Figure 8.1: Overall System Architecture

Chapter 9

Results and Discussion

The system successfully detects fraudulent transactions and records predictions on the blockchain. All stored records can be verified using blockchain explorers, demonstrating immutability.

The integration of blockchain introduces minimal latency while significantly improving transparency and trust.

Chapter 10

Limitations and Future Work

The current system uses limited transaction features and incurs blockchain transaction costs. Future extensions may include:

- Graph-based fraud detection
- Deep learning models
- Layer-2 blockchain solutions
- Privacy-preserving techniques

Chapter 11

Conclusion

This project demonstrates that combining machine learning with blockchain technology can create secure, transparent, and trustworthy fraud detection systems. The proposed framework is suitable for future decentralized financial and auditing applications.

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